Fiscal Year:	FY 2017	**Task Last Updated:**	FY 04/05/2017
PI Name:	Vanapalli, Siva		
Project Title:	Determining Muscle Strength in Space-Flown Caenorhabditis elegans		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	siva.vanapalli@ttu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	806-742-1757
Organization Name:	Texas Tech University		
PI Address 1:	Chemical Engineering		
PI Address 2:	6th Street and Canton, Mail stop 3121		
PI Web Page:			
City:	Lubbock	**State:**	TX
Zip Code:	79409-0001	**Congressional District:**	19
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	05/19/2015	**End Date:**	08/31/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 8/31/2019 per F. Hernandez/ARC (Ed., 3/5/18) NOTE: Extended to 5/18/2018 per F. Hernandez/ARC and NSSC information (Ed., 6/27/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Blawzdziewicz, Jerzy Ph.D. (Texas Tech University)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	A major impediment to long-duration space travel is loss of muscle mass and strength during spaceflight. Surprisingly, the small worm Caenorhabditis elegans is a good, cost-effective model for studying the effects of spaceflight upon muscle. Our past experiments on Earth and on the International Space Station (ISS) have shown that the expression of key muscle genes that encode members of a muscle attachment complex is reduced in C. elegans during spaceflight. A key open question is if and how these changes in gene expression result in reduced muscle strength in response to spaceflight. While reduced force production is documented for astronauts, it is yet to be demonstrated for C. elegans. To address this critical knowledge gap, we developed a tiny new device that enables continuous measurement of worm muscle forces. Using this device, on Earth, we have shown that defects in muscle attachment complexes lead to decline in worm muscle strength. Motivated by these past flight data and ground results, we propose to (a) demonstrate the flight-readiness of our miniaturized device for quantitative characterization of muscle strength of C. elegans, (b) test our hypothesis, based on past flight data, that muscle strength decreases in flight as a result of altered muscle gene expression in the worm, and (c) quantify muscle strength loss in spaceflight across multiple worm generations. Successful measurements of muscle strength in flight will make C. elegans a much more complete and comprehensive genomic model for muscle physiology, enabling meaningful translation of discoveries made in C. elegans to astronauts. Our proposed studies have important spin out implications for sarcopenia (an urgent public health problem on Earth) and may produce a terrestrial model for testing potential therapeutic interventions for muscle strength decline in flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our studies have important spin out implications for sarcopenia (an urgent public health problem on Earth) and may produce a terrestrial model for testing potential therapeutic interventions for muscle strength decline in flight.
Task Progress:	Following our suggested ground testing timeline, we have made significant progress on many aspects of our project, including acquisition of axenic media, culturing of selected nematode strains in the new media, characterization of nematode growth, strength measurements of nematodes, and development of analysis software. After reaching out to a number of companies for quotes on the custom media, C. elegans maintenance media (CeMM), we found a company, Cell Guidance Systems, able to make the media following our custom formulation at a very affordable rate relative to the cost of this custom media purchased by individuals in the past. Once NASA released funding to us, we were able to place the order and obtain the media in about 6 weeks, as is a customary turnaround time for custom media. We also kept some special order chemicals on hand with Cell Guidance Systems for any future orders of CeMM that may be necessary. Having made this connection with a reliable media manufacturing company, we have established the groundwork essential for current and future experiments that rely on use of this special media. Introducing C. elegans into this axenic media is a process met with difficulties, especially for first time users. Given that the media is extremely nutrient dense, the inadvertent addition of any bacteria or foreign microorganisms will lead to a very severe contamination that is obvious to the naked eye within a few days. The nematode must initially be transferred from its typical culture of agar and E. coli to CeMM, so cross contamination can be difficult to avoid. After some initial contamination issues, we successfully started cultures in the axenic media with the wild-type animal as well as the unc-112 and unc-52 muscular mutants suggested for use in spaceflight. The cultures are currently being maintained in 60 mm petri dishes with a media volume of 2.5 mL to allow sufficient oxygenation and minimize usage of media until BioCells become available for culture maintenance. Approximately every 2-3 weeks, a 100 µL aliquot of a crowded culture is introduced to a fresh 2.5 mL of CeMM to prevent overcrowding and starvation of a culture. As per the literature suggestions, streptomycin and kanamycin are used at concentrations of 200 µg/mL and 20 µg/mL, respectively. Wild-type animals were characterized for their growth in the media to ensure that they were developing in a manner consistent with that reported in the published literature. Literature on CeMM-grown worms reports growth data from worms on the front end of the population (i.e., those growing the most), while our data represents the average animal in the culture at the time point. Considering this slight difference, growth data and trends are consistent with those previously reported in the literature. Some animals in the population reach maturity beginning on Day 7, with the last animals becoming mature on Day 11. This wide variation among individual animals is also consistent with CeMM-cultured nematodes from previously published studies. The animals grown in CeMM are much thinner in their diameters compared to animals cultured in the standard bacterial environment. Because our NemaFlex platform is developed for worms of specific size ranges, a new device was designed and fabricated for use specifically with young adult nematodes cultured in CeMM. While the NemaFlex device used for young adult animals cultured in E. coli features a pillar edge-to-edge spacing of around 60 µm, the newly designed device has an edge-to-edge spacing of 40 µm. This tighter spacing ensures that the skinnier CeMM-cultured worms are adequately challenged by the pillars in the microfluidic environment. The closer contact of the worm with the pillar environment ensures that the animals are able to deflect the pillars rather than just moving past them with minimal contact. After development of this optimized NemaFlex platform, we obtained strength measurements for wild-type animals at various time points. These strength measurements were done on an age-synchronized population beginning at a time point where approximately half of the worms had reach maturity (Day 9 of life) through the time when all worms reached adulthood (Day 11 of life). Animals across this time period were not significantly different from one another in their strength. This indicates that strength values for worms of slightly different ages (Day 9-11) can be grouped together as a population. This becomes important in future work when only the approximate, not exact age, of a worm is known when selecting for worms from a population of mixed ages. For comparison purposes, we looked at the strength values of CeMM-cultured nematodes alongside nematodes cultured in bacteria that were at approximately the same developmental stage. Because the worms are drastically different in size, and we have previously found a cubic dependence on the diameter of the worm, we scaled the strength of the worms by d^3. Accounting for the different sizes of the worms may be necessary if worms cultured in space are smaller on average than worms cultured on Earth. Even after accounting for the size of the worm, animals cultured in CeMM are significantly weaker than their E. coli-cultured counterparts ($p<0.05$ for all time points). This could potentially be a result of minimal surface tension in the liquid axenic media, a very different environment than the solid agar with E. coli., or simply from the different nutritional source. The data has potentially interesting implications in muscle development research.

	The other area where major improvements were made was the in-house built image processing software. Our software is designed to track the deflections of the pillars that the nematode comes into contact with. Some fundamental changes were made to the algorithm that led to a >2.5x increase in processing speed of the image sequences, and some changes were made to the user interface as well to increase user friendliness. The main change to the algorithm involved the implementation of a two-step identification process of the pillars for tracking. For this, a circular Hough transform followed by grid verification improved the speed and accuracy of pillar tracking. The improvements to the user interface include easier visualization and real-time tracking of pillars being deflect by the worm. Pillars that the worm comes into contact with at some point during the duration of the image sequence are shown in blue, while pillars that are currently being deflected by the worm are circled in red. The user can therefore visualize the pillar number and location as the worm is deflecting them. This makes validation of the data during the processing step much more user friendly.
Bibliography Type:	Description: (Last Updated: 10/20/2025)
Abstracts for Journals and Proceedings	Hewitt JE, Rahman M, Van Bussel F, Blawzdziewicz J, Szewczyk N, Driscoll M, Vanapalli SA. "Measurement of muscle strength of C. elegans for space-flight studies." 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Anupom T, Rahman M, Gupta S, Vanapalli SA. "An Automated Lifespan Machine for Aging Studies in Caenorhabditis elegans on ISS and Measurement of muscle strength of C. elegans for space-flight studies." 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Hewitt JE, Rahman M, Van Bussel F, Blawzdziewicz J, Szewczyk N, Driscoll M, Vanapalli SA. "NemaFlex: A microfluidic tool for phenotyping (neuro)muscular strength in C. elegans." Texas Tech Office of Research Commercialization (ORC) Annual Meeting, Lubbock, TX, May 2016. Texas Tech Obesity Research Cluster (ORC) Annual Meeting, Lubbock, TX, May 2016. , May-2016
Awards	Hewitt JE. "2nd Place Overall Graduate Student Poster Competition Award for: Hewitt, J.E., Rahman, M., Van Bussel, F., Blawzdziewicz, J., Szewczyk, N., Driscoll, M., Vanapalli, S.A. 'Measurement of muscle strength of C. elegans for space-flight studies.' 32nd Annual Meeting of the American Society for Gravitational and Space Research. Cleveland, OH, October 2016." Oct-2016
Awards	Rahman M. "CASIS Student Investigator Flight Award in Life Sciences for Mizanur Rahman. 'An Automated Lifespan Machine for Aging Studies in Caenorhabditis elegans on ISS and Measurement of muscle strength of C. elegans for space-flight studies.' 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH. October 2016." Oct-2016
Awards	Hewitt JE. "1st Place Overall Poster for: Hewitt, J.E., Rahman, M., Van Bussel, F., Blawzdziewicz, J., Szewczyk, N., Driscoll, M., Vanapalli, S.A. 'NemaFlex: A microfluidic tool for phenotyping (neuro)muscular strength in C. elegans.' Texas Tech Obesity Research Cluster (ORC) Annual Meeting, Lubbock, TX, May 2016." May-2016